



EMPLOYEE MANAGEMENT SYSTEM



CONSOLE BASED APPLICATION IN PYTHON

Developed by

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PROJECT OVERVIEW

Employee Management System is a terminal based application built using Python. It helps to manage employee records and attendance efficiently. It supports adding, updating, searching, deleting employee details, department management, attendance tracking and salary reports. All data is stored in text files.



KEY FEATURES

- ✓ Add Employee
- ✓ View Employees
- ✓ Search Employee
- ✓ Update Employee
- ✓ Delete Employee
- ✓ Department Wise List
- ✓ Mark Attendance
- ✓ View Attendance
- ✓ Salary Report
- ✓ Total Employees
- ✓ Highest & Lowest Salary Employee
- ✓ Employee Experience

EMPLOYEE MANAGEMENT SYSTEM

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1. Add Employee
2. View Employees
3. Search Employee
4. Update Employee
5. Delete Employee
6. Department Wise List
7. Mark Attendance
8. View Attendance
9. Salary Report
10. Total Employees
11. Highest Salary Employee
12. Lowest Salary Employee
13. Employee Experience
14. Exit



Manage Employees Efficiently

Enter your choice: _



TECHNOLOGIES USED



Python
Programming
Language



File Handling
(Text Files)



OS Module
for File
Operations



SKILLS DEMONSTRATED

- Python Programming
- File Handling
- Data Management
- Problem Solving
- Logical Thinking
- Menu Driven Programming



WORKING PROCESS



User selects
an option from
the menu



Enter details /
Information



Data is saved in
text files



Records can be
searched/updated/
deleted



Reports are
generated



Exit



DATE
July 2026



DATA STORAGE
Text Files
(employees.txt,
attendance.txt)



APPLICATION TYPE
Console Based
Application



DEVELOPER
Gunjan Dahale

No.	Title	Short Statement
1.	Title	Overview of Employee Management System.
2.	Introduction	Console-based application developed in Python.
3.	Modules Used	OS module and file handling concepts.
4.	Employee Information	Employee ID, name, age, salary, email, etc.
5.	Department Management	Department-wise employee list management.
6.	Attendance	Mark and manage attendance records.
7.	Employee Operations	Add, search, update, and delete employees.
8.	Reports	Salary reports and employee statistics.
9.	Data Storage	Store records in text files.
10.	Conclusion	Project outcomes and future scope.

Technologies Used :-

Technology	Purpose
Python	Programming language
File Handling	Store employee data
Text Files	Database
OS Module	File operations

Features :-

- Add Employee
- View Employees
- Search Employee
- Update Employee
- Delete Employee
- Department-wise Employee List
- Mark Attendance
- View Attendance
- Salary Report

Employee Details Stored :-

Employee ID
 Name
 Age
 Gender
 Department
 Designation
 Salary
 Phone Number

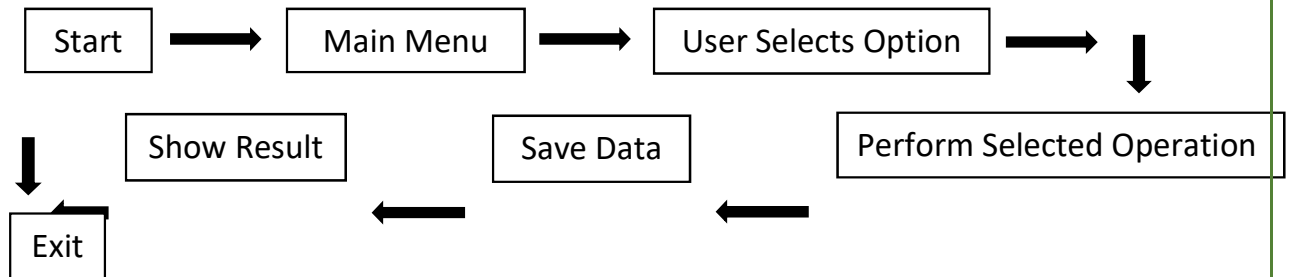
File Structure :-

- main.py
- employees.txt
- attendance.txt

Modules Used :-

os
File handling (open, read, write)

Project Workflow :-



1. Main Menu :-

```
===== EMPLOYEE MANAGEMENT SYSTEM =====
1. Add Employee
2. View Employees
3. Search Employee
4. Update Employee
5. Delete Employee
6. Department Wise List
7. Mark Attendance
8. View Attendance
9. Salary Report
10. Exit
Enter Choice: █
```

2. Add Employee :-

```
Enter Choice: 1
Employee ID: 03
Name: Ved Nagare
Age: 22
Gender: Male
Department: Finance
Designation: Manager
Salary: 40000
Phone: 8877669988
Email: vednagre56@gmail.com
Employee Added Successfully
```

2. View Employee :-

```
Enter Choice: 2
-----
Employee ID : 01
Name       : Gunjan Dahale
Age        : 21
Gender     : Female
Department : IT
Designation : Developer
Salary     : 40000
Phone      : 9988776655
Email      : gunjandahale34@gmail.com
-----
Employee ID : 02
Name       : Laxmi Dahale
Age        : 23
Gender     : Female
Department : Cyber Security
Designation : SOC Analyst
Salary     : 50000
Phone      : 8877665544
Email      : laxmidahale45@gmail.com
-----
```

3. Search Employee :-

```
Enter Choice: 3
Enter Employee ID: 02

Employee Found

Employee ID : 02
Name : Laxmi Dahale
Age : 23
Gender : Female
Department : Cyber Security
Designation : SOC Analyst
Salary : 50000
Phone : 8877665544
Email : laxmidahale45@gmail.com
```

4. Department wise List :-

```
Enter Choice: 6
Enter Department: IT
01 Gunjan Dahale Developer 40000
```

```
Enter Choice: 6
Enter Department: Cyber Security
02 Laxmi Dahale SOC Analyst 50000
```

```
Enter Choice: 6
Enter Department: Finance
03 Ved Nagare Manager 40000
```

5. View Attendance :-

```
Date : 01-08-2026  
Employee ID : 01  
Status : Present
```

6. Salary Report :-

```
Salary Report  
Total Salary : 130000  
Highest Salary : 50000 - Laxmi Dahale  
Lowest Salary : 40000 - Gunjan Dahale
```

Advantages :-

- Easy to use
- Saves time
- Reduces paperwork
- Efficient data management
- Simple interface

Conclusion :-

The Employee Management System is a terminal-based application developed using Python and file handling concepts. It helps in managing employee records, attendance, and salary reports efficiently. The project provides features such as adding, searching, updating, and deleting employee information. It demonstrates the use of programming concepts like functions, loops, conditions, and file operations. This system reduces manual work and improves data management accuracy. Overall, the project enhanced my problem-solving and Python programming skills.